

28 April 2026

Exercise 1 : SQL fundamentals

①. retrieve all columns from the employees table.

```
SELECT *
FROM employees;
```

ID	first name	last name	department	salary	hire date	city
1	Alice	Green	IT	70000	2020-01-10	JHB
2	Brian	Lee	HR	45000	2019-03-22	LT
3	Cathy	Zulu	finance	65000	2018-07-18	DBN
4	David	Makenna	Marketing	50000	2021-11-05	PTA
5	Eva	Naidoo	IT	72000	2017-09-30	JHB

② find all unique departments.

```
SELECT DISTINCT department
FROM employees;
```

Department
IT
HR
FINANCE
Marketing

③ Retrieve first and last names ordered by salary descending

```
SELECT first name,
       last name
FROM employees
ORDER BY salary Desc;
```

first name	last name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mutsona
Brian	Lee

④ Retrieve top 3 highest paid employees.

```

SELECT ID,
       first name,
       last name,
       Salary
FROM Employees
ORDER BY Salary Desc
LIMIT 3;

```

ID	first name	last name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

⑤ find employees in the IT department.

```

SELECT ID,
       first name,
       last name,
       department
FROM Employees
WHERE Department = 'IT';

```

ID	first name	last name	department
1	Alice	Green	IT
5	Eva	Naidu	IT

⑥ find employees in finance with salary > 60000.

```

SELECT ID,
       first name,
       last name,
       Department,
       salary,
FROM employees
WHERE department = 'finance' AND salary > 60000;

```

ID	first name	last name	Department	salary
3	Cathy	Zulu	finance	65000

⑦ find employees in HR or Marketing.

```

SELECT ID,
       first name,
       lastname,
       Department
FROM employees
WHERE department = 'HR' OR 'Marketing';

```

ID	first name	last name	Department
2	Brian	Lee	HR

⑧ find employees NOT in IT.

```
SELECT ID,  
       first name,  
       last name,  
       Department  
FROM employees  
WHERE department IS NOT IN 'IT';
```

ID	first name	last name	Department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	McCoena	Marketing

⑨ find employees in IT OR HR OR Finance using IN.

```
SELECT ID,  
       first name,  
       last name,  
       Department  
FROM employees  
WHERE Department IN ('IT', 'HR', 'FINANCE');
```

ID	first name	last name	Department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

(10) Find employees ~~HH~~ IN IT with salary > 65000 AND city Johannesburg.

```
SELECT ID,  
       first name,  
       last name,  
       Department,  
       salary,  
       city  
FROM employees  
WHERE Department = 'IT' AND salary > 65000 AND  
       city = 'Johannesburg';
```

ID	first name	last name	Department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg